Adobe University Hackathon 2026 · Chameleon Challenge · stage one

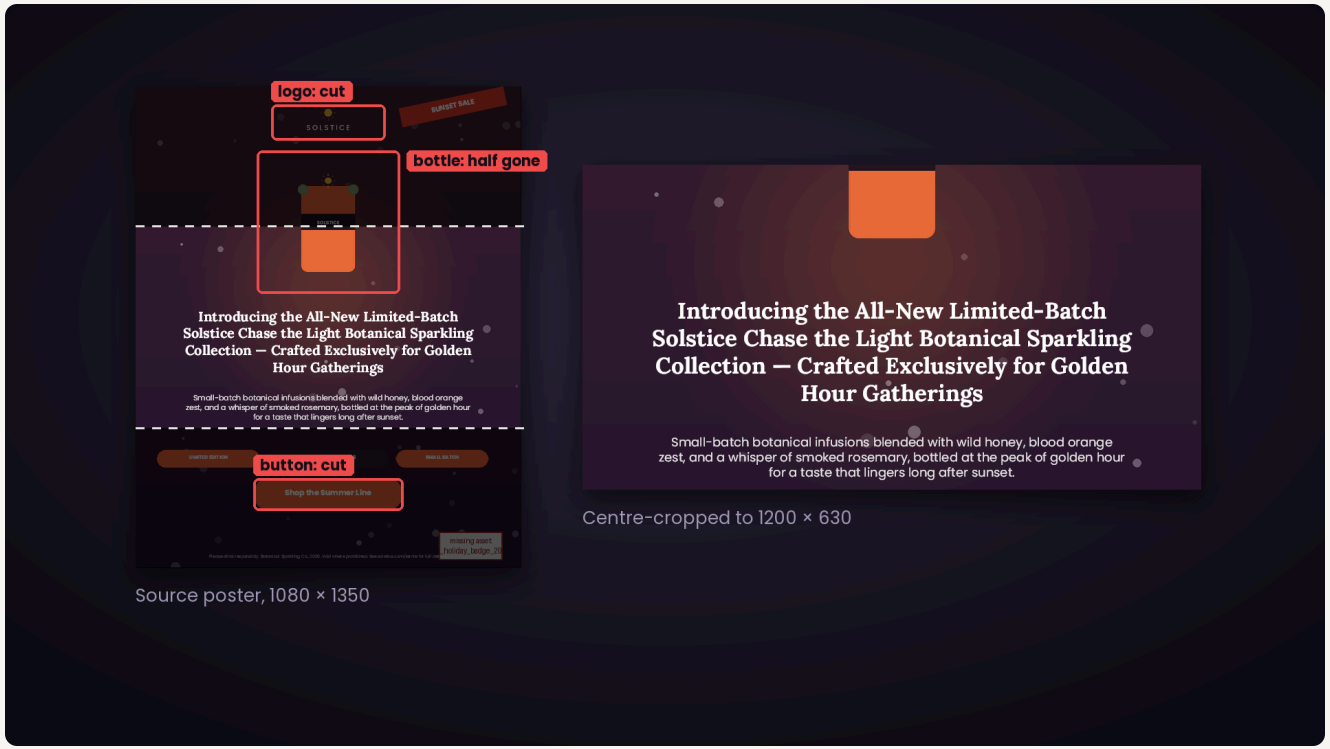
Chameleon

We built a resize engine that picks up the pieces of a design and puts them down again for the new canvas, instead of stretching or cropping the whole thing.



Chameleon, 1600 × 400

What resizing does today

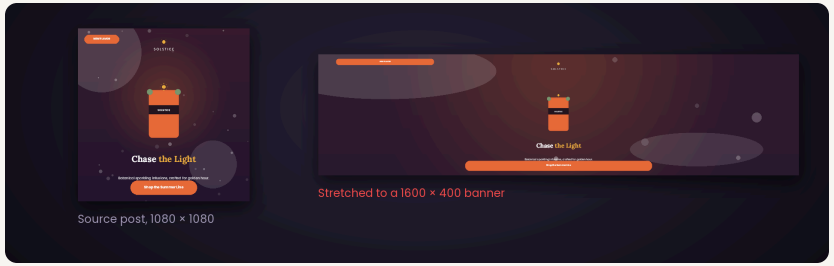


Source poster, 1080 × 1350

Centre-cropped to 1200 × 630

We started where the brief did. Take the 1080 × 1350 poster from the starter kit and ask for a 1200 × 630 Facebook post.

A centre crop loses the logo, the button and half the bottle. Stretching (below) keeps everything, but squashes it until you can't read it.



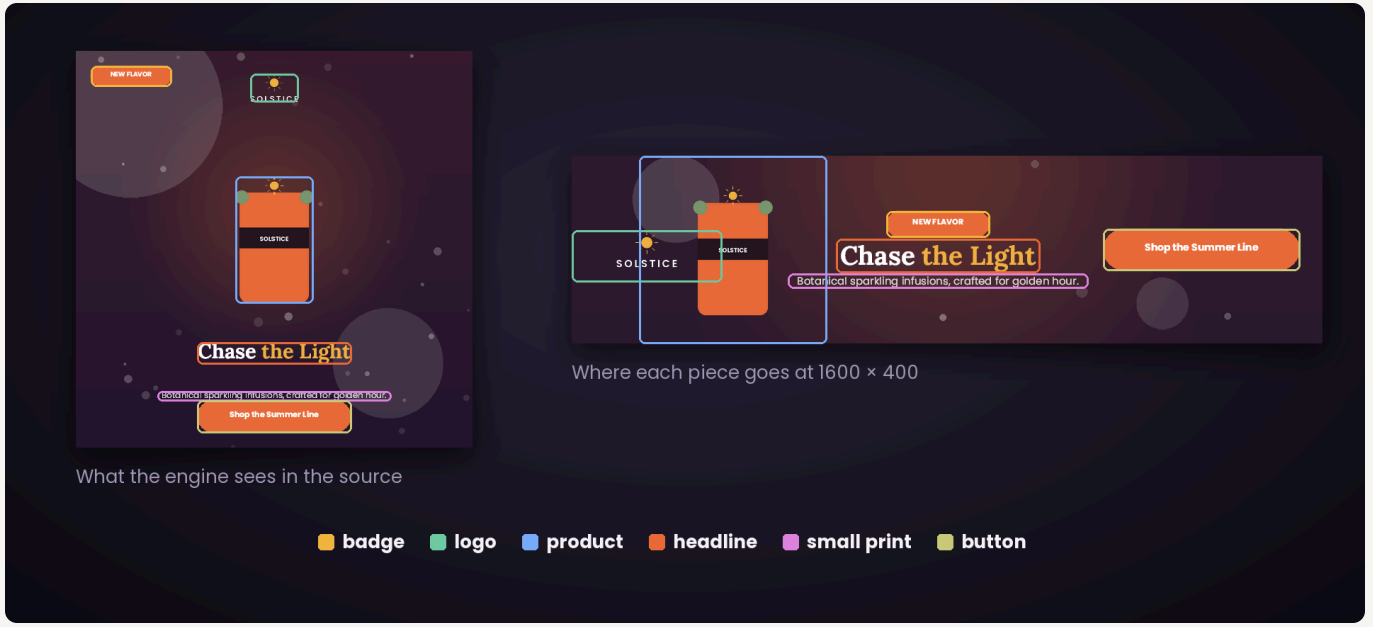
Source post, 1080 × 1080

Stretched to a 1600 × 400 banner

How it decides where things go

The input has no labels, so the engine works them out: text on a shape is a button, the biggest type is the headline, a small picture near the top is probably the logo. A badge and its label move as one piece.

Then it picks a layout for the shape and finds the largest size where every line still fits, measured with the renderer's own fonts.



Square or taller than wide	Rows, top to bottom, in the source's reading order and spacing
Landscape, up to about 2.4 : 1	Product on one side, the text column on the other (scored against rows)
Wider than that	One row: pictures, then the text stack, then the button

The hardest sample, squeezed into a 728 × 90 ad



The level 3 poster has a 150-character headline, three badges, a rotated ribbon and an image file that doesn't exist.

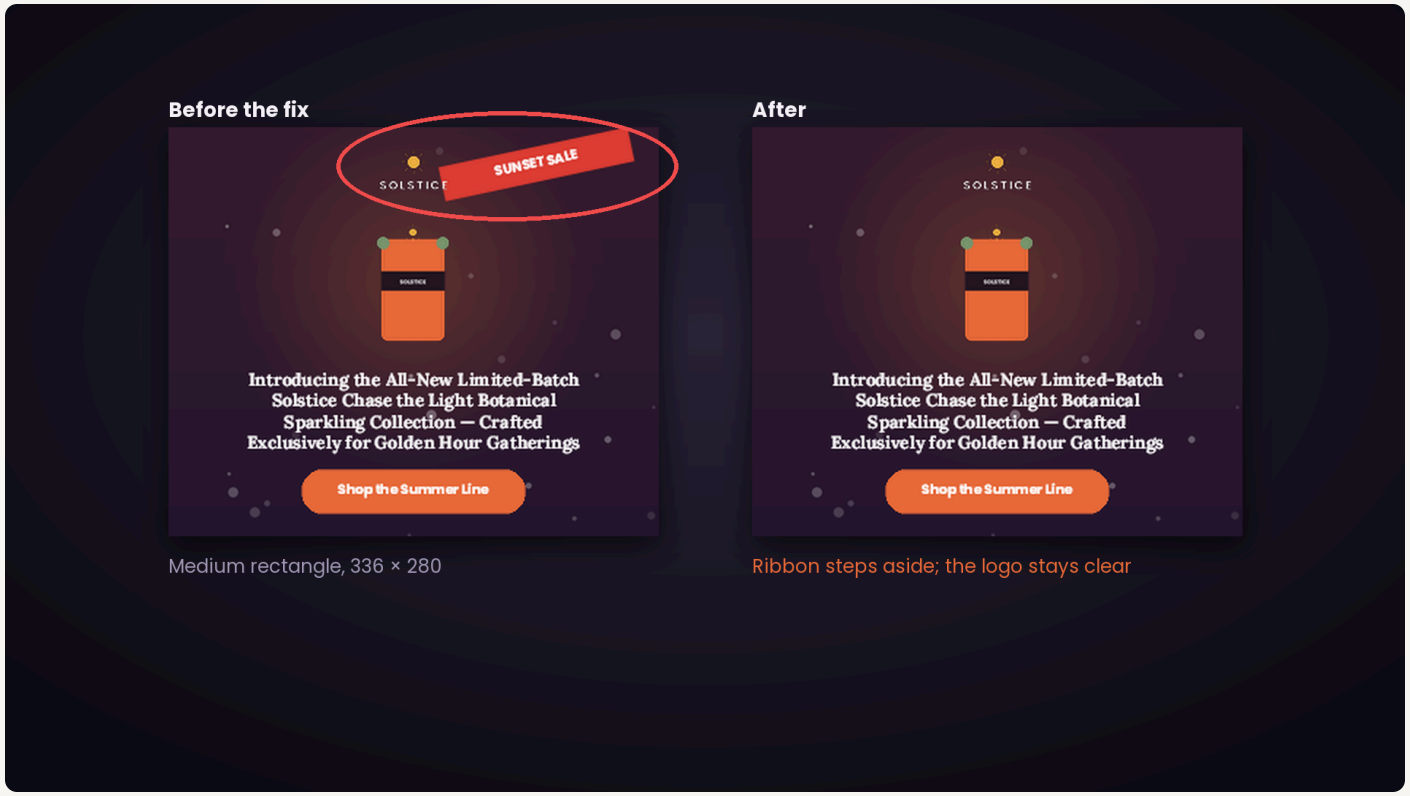
At 728 × 90 there is no room for all of it. The least important pieces go first: small print, then the ribbon, then the badges, then the subhead. Each one goes only if removing it makes the rest more readable. The logo and the headline always stay.

Removed elements are hidden (opacity 0), never deleted, so ids and counts match the source.



Leaderboard ad, 728 × 90: the badges, fine print and ribbon stepped aside

Something we got wrong



Medium rectangle, 336 × 280

Ribbon steps aside; the logo stays clear

In our last round of checks we rendered every sample at a dozen formats we had never tested. At 336 × 280 the SUNSET SALE ribbon landed right on the logo.

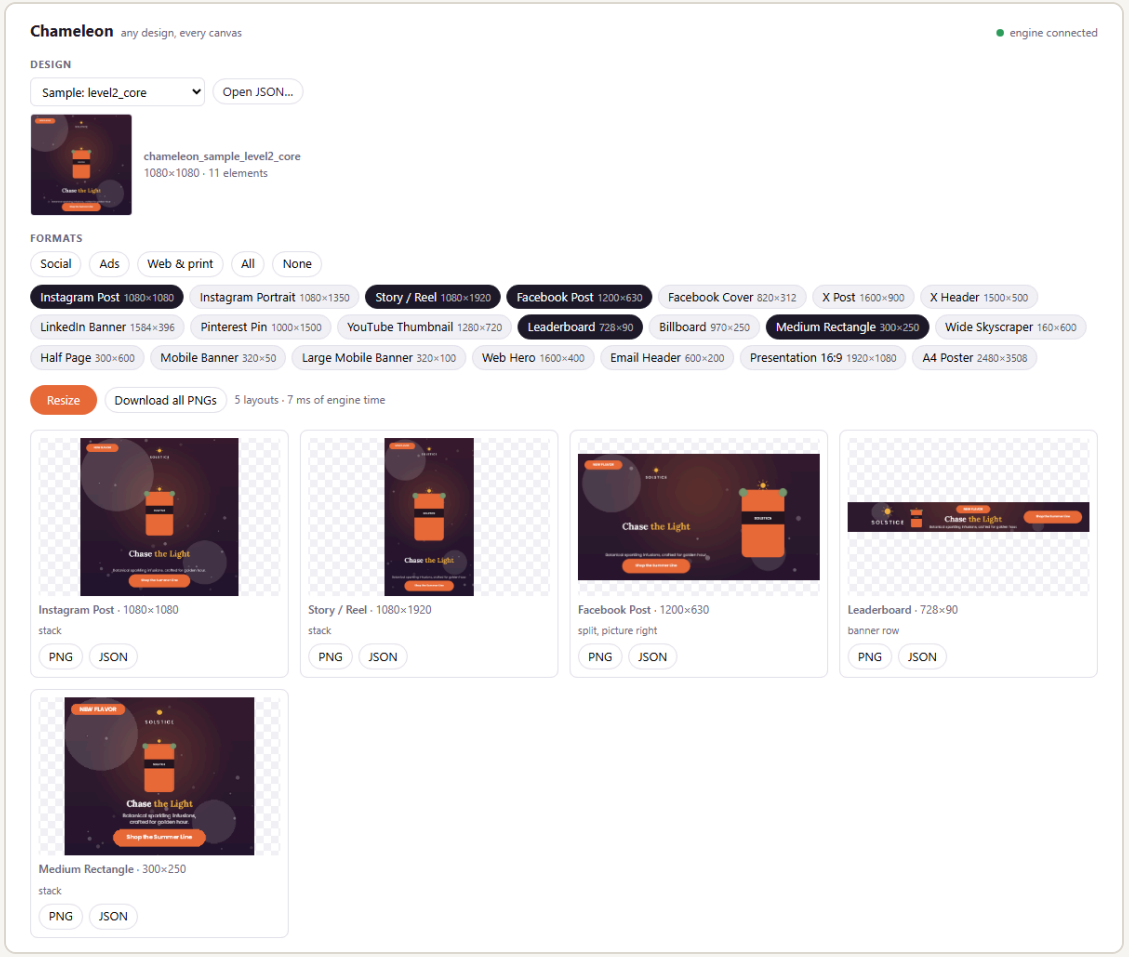
Corner stickers were allowed to overlap pictures, which is what you want for a price tag on a bottle. But the logo is a picture of a word.

The fix was one condition: stickers now treat the logo like text. When there's no room, the ribbon steps aside instead.

How we check it

3 official samples and 7 stress designs, each at 20 real formats	200 resizes. Every hard rule is asserted: ids, in-canvas (rotation included), image proportions, unchanged copy
Does the text really fit?	Checked with the renderer's own line wrapping, so "fits" means fits in the PNG, not just in our maths
1,500 randomly generated designs	No crashes, no rule violations, no fallbacks
Speed	1 to 150 ms per layout. Deterministic, offline, no model
The Docker image	Built for linux/amd64 and run with `--network none` in CI on every push, like the scorer
Pixels generated	None. Only positions, sizes and font sizes change

In Adobe Express



The add-on runs the same engine on the page you're editing. Pick some formats and press Resize. Each one is added as a new page with real text and shapes you can keep editing.

Anything the engine set aside is hidden on that page, not deleted, so you can bring it back.

Each result says what it did, for example "banner row · 3 secondary elements set aside".

What we'd build next

1. Show the best three layouts for each format. The engine already scores the candidates and only shows the winner.
2. Let people lock an element in place and lay everything else out around it.
3. A small local vision model for focal points when cropping photos, and as a second opinion on roles.
4. Shorter, approved copy for tiny ad slots. Stage one rules this out on purpose, but it's where the biggest remaining gains are.

Engine, add-on, tests and this deck: github.com/UditDewan/chameleon-adobe-hackathon